

Zachariah Loy Yiqi

Singaporean

✉ loyzachariah@gmail.com

☎ (65+) 91860874

 LinkedIn: <https://www.linkedin.com/in/zachariah-loy-84a083274/>

 GitHub: <https://github.com/trashprog>

SUMMARY

Zachariah is an aspiring data engineer pursuing a diploma at Ngee Ann Polytechnic, combining strong academic performance with practical experience in environmental tech and data analysis. Proficient in Python, SQL, and various data tools, with active involvement in AI and software development communities.

EDUCATION

Singapore

Ngee Ann Polytechnic

School of Infocom Technology | Diploma in Data Science | Minor in Internet of Things

Expected Graduation, May 2026

- **Concentrations:** Data Analytics, Machine learning, Data Engineering
- **CGPA:** 3.92/4.00, *Director's List*
- **Related Coursework:** Data Exploration and Analysis, Machine Learning, Artificial Intelligence, Object-Oriented Programming, Statistics & Applications, Databases, Agentic Chatbot Development, Retrieval Augmented Generation (RAG), Deep Learning.

EXPERIENCE

Singapore

Center for Climate Research Singapore

AI-NWP Intern

Sep 2025 – Jan 2026

- Benchmarked 5 AI-NWP models using NRMSE (Normalized Root Mean Squared Error) based on traditional and dynamic metrics as well as the Radical Average Spectrum.
- Wrote python and bash scripts to download, process and run inference on ECMWF Operational Analysis data for model evaluation and ran them on HPC systems.
- *Co-authored a research paper with my supervisor based on visualizations plotted from computed results (2nd Authorship).*

Center for Environmental Sustainability

Data Analyst

Oct 2024 – Feb 2025

- Cleaning project files with power query editor and suggested improvements to storing the data Created Power BI Dashboard for Admin Senior Manager
- Creating dashboards to track Project leader and member performance

Product Tester and Programmer

Aug 2024 – Oct 2024

- Exhibit Panel User Interface (Nextion)
- Created Power BI Dashboard for Admin Senior Manager
- Utilised Arduino for lighting using ZigBee (Mesh)

PROJECTS

Agentic Compost Chatbot (Capstone Project)

April 2025 – August 2025

- Built a chatbot application using Docker, PostgreSQL, LlamaIndex, AWS S3, FastAPI, ExpressJS, and React.
- Implemented user management features, including authentication, persistent chat history, and image storage.
- Designed an agentic workflow to dynamically route queries:
 - Text-to-SQL for live IoT database queries

- Retrieval-Augmented Generation (RAG) for domain-specific composting questions
- Enabled real-time, streaming chatbot responses for improved user experience.

Real-Time Facial Augmentation Pipeline (Computer Vision Project)

April 2025 – August 2025

- Built a real-time CV pipeline using Python, OpenCV, and MediaPipe to apply AI-generated facial texture augmentations on live video.
- Generated synthetic facial textures (aged skin, jaundice, smoker lines) using a local generative AI model to expand limited datasets.
- Selected for Project ID: Industry Showcase (Top 8 / ~500 teams).

ELT Pipeline (Agile DataOps Project)

Oct 2024 - Feb 2025

- Built a fully automated ELT pipeline integrating AWS S3 and Snowflake, enabling hands-free ingestion of CSV files via Snowpipe auto-ingest. Made using streams, tasks, and dynamic MERGE statements
- Implemented schema inference and dynamic table creation, allowing new datasets to be onboarded without manual DDL.
- Designed MD5-based change detection and incremental loading logic, ensuring only modified files and records are merged into existing tables.

Bevy Rust Game

Space Game – Cell War

Feb 2024 – April 2024

- Leveraged Bevy's Entity Component System (ECS) to Create various entities in the game.
- Applied the knowledge of vectors from Sem 1.2 module Mathematics for data science to code out physics of the bullet shots.
- Created 8 enemy types with different behaviors.

ACTIVITIES AND AWARDS

Certifications

- [AWS Certified Cloud Practitioner](#) - AWS
- [AI Agents and Fundamentals](#) – Hugging Face
- [PSM1](#) – Scrum.org
- [Certificate of Excellence](#) – School of Humanities and Interdisciplinary Studies – Ngee Ann Polytechnic

Overflow Special Interest Group (SIG)

Workshops attended

Oct 2023 – Dec 2023

- Data Structures and Algorithms bootcamp 2023
- Building personal portfolio website with Figma and web development
- Docker, concepts, Dockerfiles, pushing and pulling with Docker hub

SKILLS

Programming: Python, SQL, C#, Javascript, Bash

Tools: Power BI, AWS, Jupyter, Agile, Snowflake, vscode, SQL Server Management Server, Excel, Arduino, SAP GUI/Cloud Analytics, Ollama, Vim, Linux, Docker, PostgreSQL, Groq, Grafana